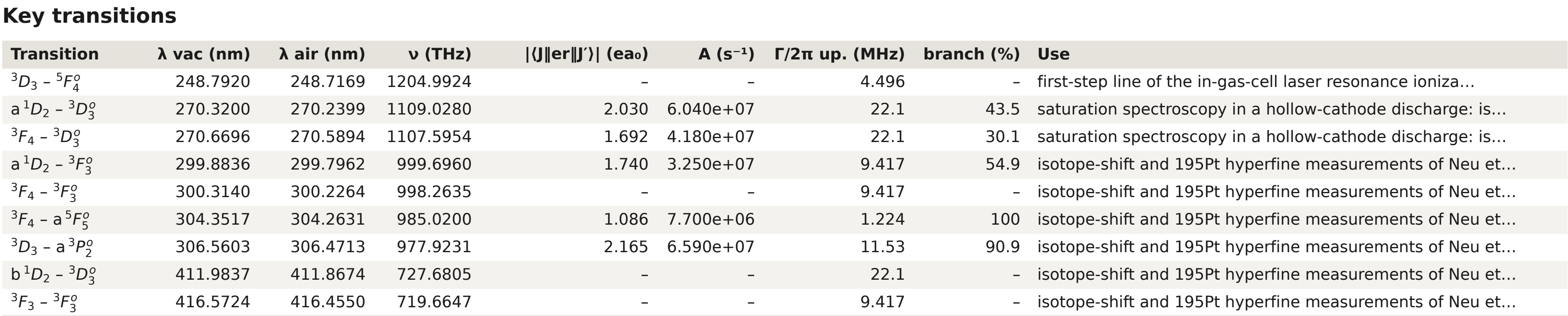
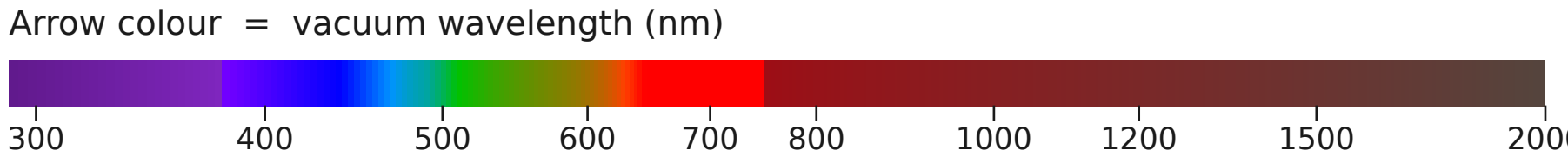


Pt⁺ ionisation limit 72257.80 cm⁻¹ = 8.95883 eV (138.393 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J||er||J\rangle|$ in ea_0
 Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ model calculation.
 Arrow width: grows with the Einstein coefficient A . Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
 Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST
 $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
3D_3	5702.64(23)	-	F=5/2: -11405.28 F=7/2: +8553.96
a^1D_2	-2607.035(4)	-	F=3/2: +3910.55 F=5/2: -2607.03
3F_4	849.0140(21)	-	F=7/2: -2122.53 F=9/2: +1698.03
3D_2	2087(15)	-	F=3/2: -3130.50 F=5/2: +2087.00
3F_3	1223(18)	-	F=5/2: -2446.00 F=7/2: +1834.50
b^1D_2	5546(15)	-	F=3/2: -8319.00 F=5/2: +5546.00
1G_4	1289(60)	-	F=7/2: -3222.50 F=9/2: +2578.00
$a^3P_2^o$	1850(5)	-	F=3/2: -2775.45 F=5/2: +1850.30
$a^3F_3^o$	3793(4)	-	F=9/2: -11379.90 F=11/2: +9483.25
$^3F_3^o$	3074(4)	-	F=5/2: -6147.60 F=7/2: +4610.70
$^3D_3^o$	1917.2(9)	-	F=5/2: -3834.40 F=7/2: +2875.80
$^3P_2^o$	1150.1(10)	-	F=7/2: -2875.00 F=9/2: +2300.00

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$^3D_3 - ^3F_3^o$	248.792	198-194	2430(2030)
$^3D_3 - ^3F_3^o$	248.792	196-194	1390(270)
$^3D_3 - ^3F_3^o$	248.792	192-194	-1430(2070)
$a^1D_2 - ^3D_3^o$	270.320	196-194	-2068(3)
$a^1D_2 - ^3D_3^o$	270.320	192-194	2005.3(24)
$a^1D_2 - ^3D_3^o$	270.320	198-194	-4312.2(24)
$a^1D_2 - ^3D_3^o$	270.320	195-194	-952.7(18)
$^3F_4 - ^3D_3^o$	270.670	190-194	8616.0(015)
$^3F_4 - ^3D_3^o$	270.670	192-194	4530.8(015)
$^3F_4 - ^3D_3^o$	270.670	196-194	-4671.1(21)
$^3F_4 - ^3D_3^o$	270.670	198-194	-9741.8(21)
$^3F_4 - ^3D_3^o$	270.670	195-194	-2137.5(15)
$a^1D_2 - ^3F_3^o$	299.884	194-192	-1631.36
$a^1D_2 - ^3F_3^o$	299.884	195-194	-782.2(27)
$a^1D_2 - ^3F_3^o$	299.884	196-194	-1692(13)
$a^1D_2 - ^3F_3^o$	299.884	198-194	-1823(13)
$^3F_4 - ^3F_3^o$	300.314	194-192	-4154(11)
$^3F_4 - ^3F_3^o$	300.314	195-194	-1956.1(21)
$^3F_4 - ^3F_3^o$	300.314	196-194	-4270(11)
$^3F_4 - ^3F_3^o$	300.314	198-192	-4634(12)
$^3F_4 - a^3F_3^o$	304.352	194-192	-2525(9)
$^3F_4 - a^3F_3^o$	304.352	195-194	-1193(9)

$$\text{Shift} = \nu(\text{first isotope}) - \nu(\text{second isotope}).$$